PlotStudio

Overview

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QUESTION

How do monthly promo-offer subscribers compare to other monthly and annual promo-offer subscribers in churn rate and customer characteristics, and what actionable business hypotheses follow from the observed and adjusted analysis?

Overview

I built an executive-ready customer churn report using the verified segment-analysis outputs and a validation check against the cleaned subscriptions data. The report confirms that **monthly promo-offer subscribers are the highest observed-churn segment**, and that this ordering remains after adjustment for overlapping observed variables. **Figure 1**

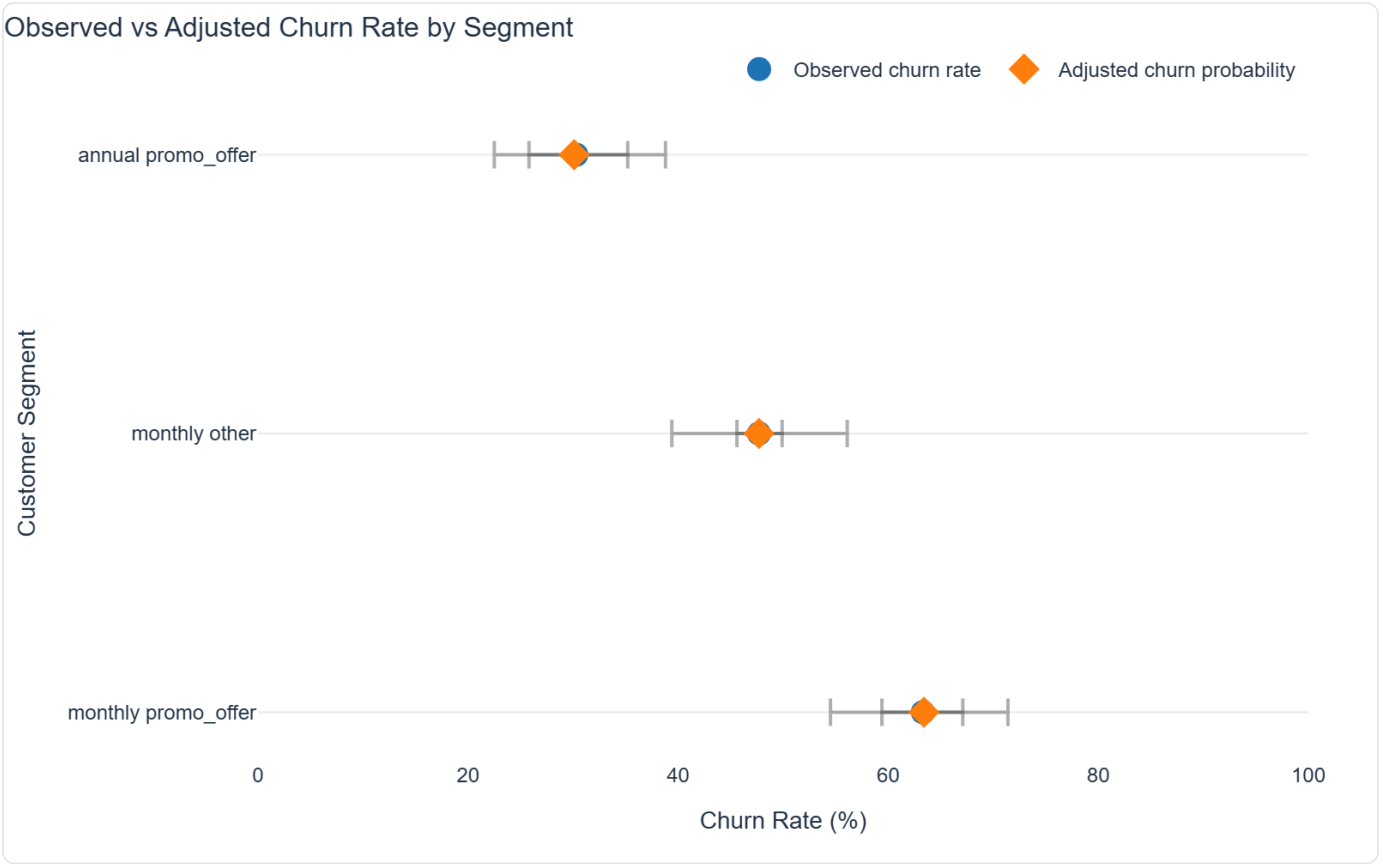


Figure 1

Key Findings

The segment comparison table `exec churn report segments` summarizes the three verified groups used in the report.

Segment	Sample size	Observed churn	Observed 95% CI	Adjusted churn	Adjusted 95% CI
Annual promo-offer	360	30.3%	25.8–35.2	30.0%	22.5–38.8
Other monthly	2,072	47.7%	45.6–49.9	47.5%	39.4–56.1
Monthly promo-offer	589	63.3%	59.4–67.1	63.0%	54.5–71.4

Key executive KPIs from the verified analysis are below.

KPI	Value
Total subscriptions	4,200
Churned subscriptions	1,768
Overall observed churn rate	42.1%

The main segment gaps are large in magnitude and directionally stable after adjustment.

Comparison	Observed interpretation
Monthly promo-offer vs other monthly	Higher observed churn by 15.6 percentage points (63.3% vs 47.7%)
Monthly promo-offer vs annual promo-offer	Higher observed churn by 33.0 percentage points (63.3% vs 30.3%)
Adjusted pattern	Monthly promo-offer remains highest after adjustment; annual promo-offer remains lowest

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Interpretation

This is **observed churn analysis, not validated future-churn prediction**. The adjusted comparison controls for overlapping observed variables from the prior verified work, but it does not establish causation. The evidence supports prioritizing the **monthly promo-offer segment** for retention investigation because it is both materially larger than the annual promo-offer group and substantially higher churn than either comparator after adjustment.

An executive-ready summary of the verified findings is:

Section	Evidence-based summary
Executive summary	Monthly promo-offer subscribers show the highest recorded churn, and that elevated risk persists after adjustment for observed demographic, device, plan-tier, and signup-cohort overlap.
Key KPIs	4,200 subscriptions, 1,768 churned, 42.1% overall observed churn.
Monthly promo-offer segment comparison	The three verified segments are annual promo-offer (n=360), other monthly (n=2,072), and monthly promo-offer (n=589), with observed churn rates of 30.3%, 47.7%, and 63.3%, respectively, each with reported 95% confidence intervals.
Adjusted analysis	The adjusted probabilities closely track the observed ordering: 30.0% for annual promo-offer, 47.5% for other monthly, and 63.0% for monthly promo-offer.
Three actionable business hypotheses	(1) Promo-acquired monthly subscribers may be arriving with weaker fit or commitment than other monthly subscribers; (2) the monthly billing structure may amplify the churn risk associated with promo-acquired customers; (3) the high-churn segment may be compositionally concentrated in acquisition or onboarding mixes not fully captured by the available variables.
Recommended experiments	(1) Randomized onboarding or early-lifecycle retention treatment within monthly promo-offer subscribers; (2) randomized promo framing or offer-eligibility test that keeps price constant where operationally possible; (3) channel-specific holdout test comparing promo-offer acquisition paths against non-promo acquisition for otherwise similar monthly plans.
Limitations	The analysis intentionally excluded `churn_date` and `cancel_reason` from predictors to prevent leakage. Results are based on observed cross-sectional churn in this extract, use verified confidence intervals, and lack a clear observation cutoff for fully comparable tenure exposure. The analysis does not validate future churn timing or intervention effects.
Conclusion	Decision-makers can reasonably treat monthly promo-offer subscribers as the top observed churn-priority segment in this dataset, while keeping interpretation descriptive and testing business responses experimentally before acting at scale.

Data Sources

- df_exec_churn_report_segments